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Train delay prediction in Tunisian railway through LightGBM model

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Abstract

Train delays are one of the most important problems in the railway systems across the world, which urges the development of predictive analysis-based approaches to estimate it. With the advanced big data analysis and machine learning tools and technologies, the train delay prediction systems can process and extract useful information from the large historical train movement data collected by the railway information system. Accurate prediction of train delays can help train dispatchers make decisions through timetable rescheduling and service reliability improving. We propose a machine learning model that captures the relationship between the arrival delay of passenger trains and various characteristics of the railway system. We present the first application of lightGBM regressor based on optimal hyperparameters to predict train delays. The proposed model performance is compared with different models that have been frequently used for such problems. To evaluate our model, we have compared its R2, RMSE, RME with that of support vector machine, random forest, Xgboost, and neural network. Statistical comparison indicates that the LightGBM outperforms and is faster than all the other tested models.

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1. Introduction

Train delay is a critical problem in the Tunisian railway systems because trains are frequently delayed, and passengers have to face much inconveniences. These delays are caused by various reasons ranging from disruptions in the operations flow, infrastructure problems such as construction work, repair work, accidents, and weather

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conditions. A late train is strongly likely to accumulate more delays or to propagate its delay with other trains. Managing these delays (rescheduling), allows traffic managers to redirect trains in order to use the rail network in a better way. Delays prediction is one of the challenges of rail systems worldwide to improve traffic management and dispatch. It consists in establishing the relationship between the delays of the trains and various characteristic of the railway system. If we can predict train delays in advance, railway operators can schedule a suitable timetable. This will minimize delays and prevent the problems in the railway plan and it will be of great help for passengers to plan their itinerary according to their own work.

Developing a highly accurate LightGBM-based passenger train delays prediction is the aim of our research. To evaluate the proposed model, a comparison with the other models has been made using the following metrics: R^2 , RMSE, MAE. The evaluation revealed that the proposed model has the highest accuracy and the minimum time-running.

In this work the sections are organized as: in the second section we discussed and showed the previous researches based in machine learning for passenger train delay prediction in Table 1. Section 3 presented our methodology with a detailed explanation for each step. In addition, we detailed the evaluation of the proposed methodology and compared obtained results with other tested models. And we concluded our study in the 5th section adding our future perspectives.

2. Literature review

With the increase of new methodologies of machine learning and artificial intelligence, that is applied in different areas and have solved numerous problems, the railway transportation systems show big interest towards these technologies to collect, process, and analyze large amounts of data and extract useful information. Machine learning methods have been widely used in literature to address train delays prediction. In this work we focused on the recent research that have used machine learning models to address passenger train delay prediction.

The first application of support vector regression (SVR) to analyze train arrival delays on Serbian Railways was presented in [1], in this case study the statistical comparison indicates that the SVR technique outperformed the ANN algorithm. A statistical analysis was applied in [2], for an accurate prediction of the lengths of running and dwelling times, using three global predictive model methods namely, Robust Linear Regression (LTS), Regression Trees (RT) and Random Forests (RF), beside local models that presented for a particular train line, station or block section, based on LTS with some refinements. These models are tested using delay history data from Rotterdam and The Hague in the Netherlands. The k-nearest neighbor (K-nn) model was applied in [3] to estimate the off-peak-hour dwell times, additionally to the Linear Regression (LR) model to predict the peak hour dwell times using data from Dutch railway stations. Three different algorithms are used by [4] to address train delay prediction problem, i.e. Extreme Learning Machines (ELM), Kernel Regularized Least Squares (KRLS) and Random Forests (RF), using data provided by Italian railway system and weather information. The performance comparison indicate that RF consistently performed better. In their more recently work [5] they proposed a neural network method called Deep Extreme Learning Machines to build a dynamic data-driven train delay prediction system. An ensemble of predictive algorithms based on artificial neural networks and genetic algorithm were proposed in [6] for train arrival time prediction namely Back-Propagation Neural Network (BPNN), Wavelet Neural Network (WNN), BPNN optimized by Genetic Algorithm (GA-BPNN) and WNN optimized by Genetic Algorithm (GA-WNN), The results show that the GA-BPNN is more efficient than other tested models. Predict the running time, the dwell, the train delay, and the penalty costs were the focus of work in [7], where a hybrid approach that combined the Decision Tree (DT) and Random Forest regression (RF) was proposed. Neural Network (NN) model was the most accurate method where it was applied and evaluated using decision tree model with and without adaboost in [8]. Several machine learning models are applied in [9], [10] and [11] to predict train delay. The comparison of obtained results in these researches indicated that the random forest outperformed the other evaluated methods. Researchers in [12] addresses the problem of forecasting train delays up to 24 h in advance, on proposing a data driven method that combines simulation and statistical approaches in an ensemble. The proposed method has tested using a very large data from the train network of Germany, where the result indicates that the method based on ensemble outperformed the component models. The study in [13] has proposed a Long Short-Term Memory (LSTM) model to predict the train arrival delay. Its performance was compared with that of RF and NN where it has exceeded them. A coupled classification–regression model was proposed in [14], where a bi-level random forest is formed of a random forest classifier in the first level to predict whether a train delay

will increase, decrease, or remain unchanged, and a random forest regressor to estimate delay in minutes given the predicted delay class at primary level. The first application of Gradient Boosting Regression Tress (GBRT) model to predict train delay was presented in [15], and the results demonstrate that the proposed model based in GBRT has a higher prediction precision and outperforms the SVR and the RF models. CLF-Net is a deep learning (DL) approach that combines 3-Dimensional Convolutional Neural Networks (3D CNN), Long Short-Term Memory (LSTM) recurrent neural network, and Fully-Connected Neural Network (FCNN) architectures, it was developed in [16] to predict train delay of two high-speed rail (HSR) lines in China, namely, the Wuhan-Guangzhou HSR and the Xiamen-Shenzhen HSR.

Table 1. A summary of literature review.

Work	Year	Models	Country of data	Results
[1]	2015	SVR	Serbian Railway	SVR outperforms the ANN
[2]	2015	LTS, RT, RF	Netherlands railway	The local models perform best both in accuracy and computation time.
[3]	2016	LR, K-nn	Dutch railway	The estimation accuracy was 85.8%–88.5% during peak hours and 80.1% during off-peak hours.
[4]	2016	ELM, KRLS, RF	Italian	Random Forest consistently performed better
[6]	2017	BPNN, WNN, GA-BPNN, GA-WNN	China	The simulation results show that the GA-BPNN outperform the competing algorithms.
[7]	2018	Hybrid approach DT + RF	Italian railway	This hybrid model performed better than either.
[8]	2018	DT with AdaBoost, DT without AdaBoost, NN	Sweden	Neural network was the most accurate method
[9]	2019	LR, SVR, ANN RF	China	Random Forest Regression (RFR) model outperforms other three alternative models.
[10]	2019	LR, GBRT, DT, RF	Indian railways	Random forest outperforms other methods
[11]	2020	RF, ANN, GBDT XGBoost	Dutch railways network	The random forest model based in optimal parameters exhibits high prediction accuracy
[12]	2019	Ensemble approaches	Germany	Ensemble method outperformed the component models
[13]	2019	LSTM, RF, NN	Dutch railway	The results indicate that the LSTM model outperforms other models.
[14]	2019	Bi-level random forest	Netherlands' railway	The bi-level model provided the most accurate predictions
[15]	2020	GBRT, SVR, RF	China	The GBRT model has a higher prediction precision and outperforms the SVR and the RF models.
[16]	2020	3D CNN, LSTM, FCNN	China	CLF-Net outperforms conventional machine learning models and the state-of-the-art DL models

Different machine learning methods have been widely used in train delay predictions. However, the outperformed model differs from one study to another depending on the data case used.

Despite the importance of this problem, it was not previously addressed in the Tunisian railway system. Therefore, our work presents the first application of machine learning to predict passenger train delays in Tunisian railway.

3. Proposed methodology

We proposed a train delay prediction in Tunisian railway system through machine learning techniques. This is the first work that aims to predict train delays in the Tunisian railway system and presents the first application of LightGBM algorithm to predict train delays. The proposed methodology is shown in Figure 1. The following steps present our process for train delay prediction using machine learning algorithms.

- Data Collection
- Data Cleaning

- Exploratory Data Analysis & Feature Engineering
- Modeling
- Evaluate Models

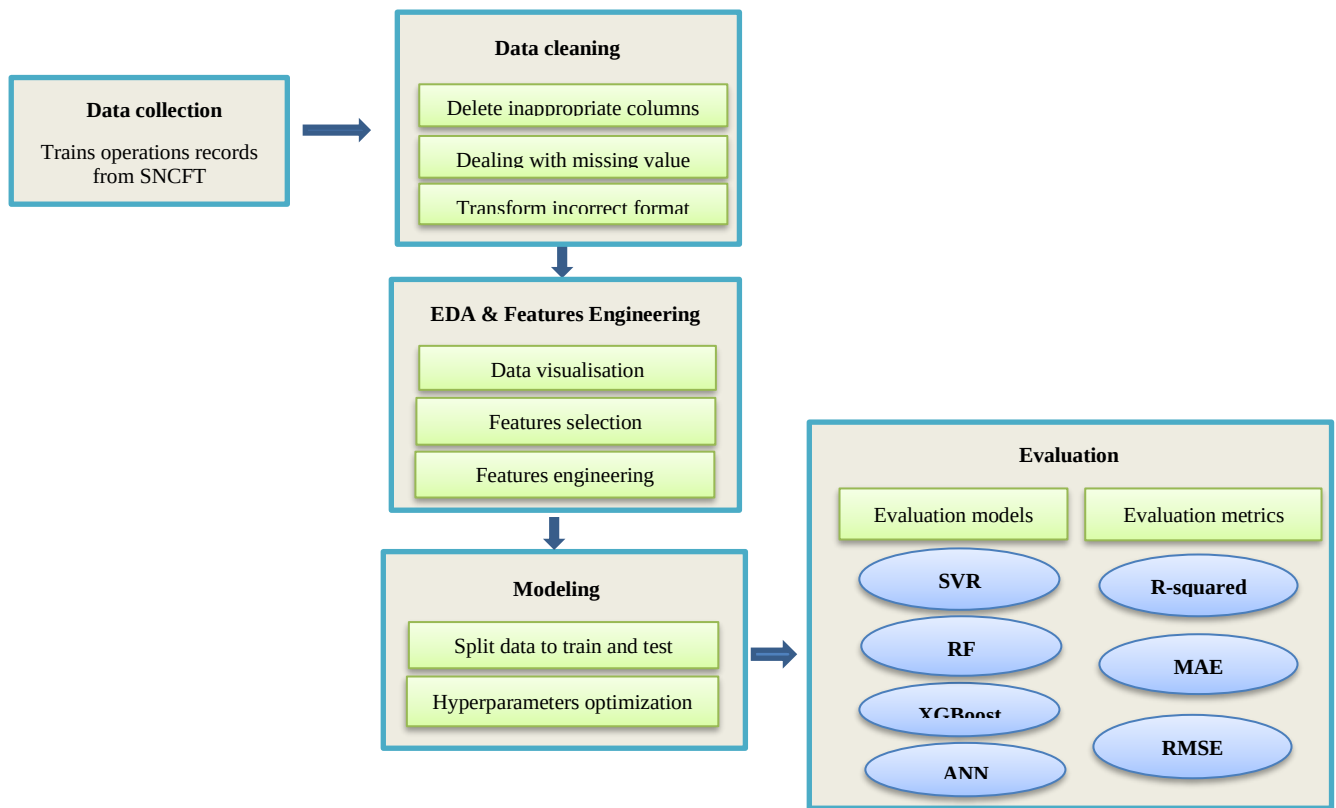


Fig. 1. Proposed approach for train delay prediction

3.1. Data collection

The Tunisian rail network comprises 23 lines, with a total length of 2165 km. In addition, the railway network includes 267 stations and stops. It has also 4 road-rail links with the aim of promoting bimodal passenger transport by combining rail and road transport. 316 trains run on the rail network daily including 246 passenger trains. The maximum speed of trains is 130 km/h on metric track and 140 km/h on standard track[†].

The database used consists of 12350 travel samples, from 1.1.2019 to 31/12/2019, registered by the SNCFT (National Tunisian Railway Company). This includes 55 passenger trains. This data includes 4 main destinations: Tunis-Nabeul (76.3 km, 26 stations), Tunis-Sfax (269.249 km, 41 stations), Tunis-Sousse (162.2 km, 32 stations), Tunis-Tozeur (234.1 km, 58 stations). The Data have the following features:

Trains info:

- Train : unique code of each train
- Line: the railroad taken by the train
- Direction: indicate if the railroad is a single-track or a double-track.
- Code_cir: How often a train is running (daily, only Saturday and Sunday, ...)

[†]<http://www.sncft.com.tn/>

Infrastructure info:

- Destination: departure station to target station
- Distance: the traveled distance
- Nbr_station: number of stations and stops between the depart and the target

Calendar info:

- Date: date of travel
- Weekday: day of travel (for Monday to Sunday)
- Holiday: a boolean variable that indicate if the journey is a holiday or not
- Month: month of travel (from January to December)
- Season: season of travel (Winter, Spring, Summer, Autumn)

Delay info:

- Motifs: the reason behind the delay
- Departure delay: how much time (in minutes) a train took to begin its new journey after the scheduled departure time.
- Departure time: the time when the train departed
- Arrival delay: how much time (in minutes) a train took to arrive after the scheduled arrival time (Our target variable).
- Arrival time: the time at which the train arrived

Data cannot come in the way we want it to be, data comes and has a lot of problems, which limits the efficiency and performance of machine learning algorithms. A cleaning step is necessary and inevitable to improve data quality before model training.

3.2. Data cleaning

This phase is very important to improve data quality and model performance. The purpose of data cleaning is to remove the anomalies that affect the model. It aims mainly at discovering data problems and solveing it. It includes dealing with null values, handling outliers and transforming incorrect format and last but not least duplicate records elimination. To deal with null values we have dropped the columns that have null values to more than 30% (only one column: Motifs) and the records that have null value in Arrival_delay column because we can't train our model on a fake data. The remaining null values have been filled either with the median (as for outliers' existence as observed in Figure 2) or with "Missing" value if the feature was categorical. Then we transformed the data type of departure delay from object to numeric. Finally, the data had no duplicates, so the cleaning stage has finished and we are ready for exploration .All cleaning operations was executed utilizing the Python libraries.

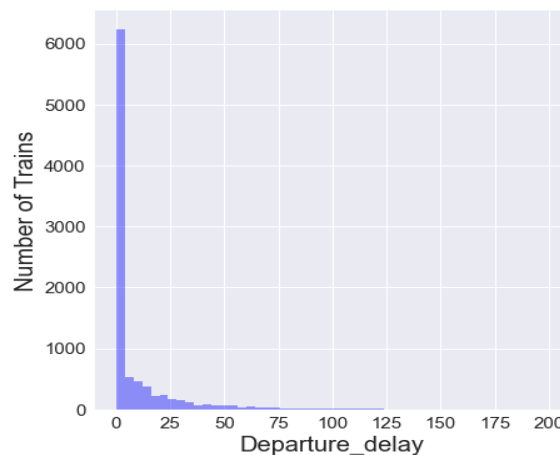


Fig. 2. Departure delay histogram (Most of the trains are leaving on time which makes the distribution skewed to the right)

3.3. Exploratory data Analysis, data Visualization and features Engineering

It is important to find the influencing features that affect train delay. However, for a better understanding of the data, a data visualization is applied using Python library, such as 'matplotlib'. We have analyzed the relationship between the arrival delays and the rest of the variables of the collected datasets. In our work we have analyzed the data in different manners. First, the univariate analysis that consists of analyzing a single variable from the dataset was used to study the distribution of variables. For example figure 3 shows the distribution of weekday data where we can observe that there is a balanced distribution. Then, if we analyze data by taking two columns into consideration from a dataset, it is known as bivariate analysis. This analysis can be with two numeric, as the figure 4 (b) that present the relationship between Departure_delay and The Arrival_delay shows. Obviously, there is a strong correlation between arrival delay and departure delay which is reasonable (the longer the trains are late in departure, the more delayed they will arrive). Or into numeric and categorical, e.g. figure 4 (a) indicates the relation between arrival delays and seasons. As we can see when It's summer the Arrival_delays tend to be relatively large (the average of Arrival_delays in Summer and Autumn is bigger than that of the other seasons). Finally, we analyzed the data by taking more than two columns into consideration, which is known as Multivariate Analysis. To see if the relationship between Arrival delays and Departure_delays differ according to whether or not it is a holiday, figure 5 was implemented. As it shows there is a difference between the density of delays when the journey is a holiday and when it is not. Logically, in normal weekdays the delays density increases due to the number of passengers and the number of running trains in same line. We have applied the same analysis for all features.

After a full analysis the parameters that were likely to have impacts on future train delays were selected as the model features (inputs(X)).

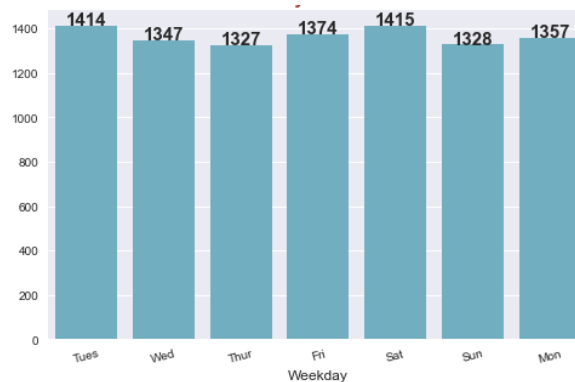


Fig. 3. Weekday distribution

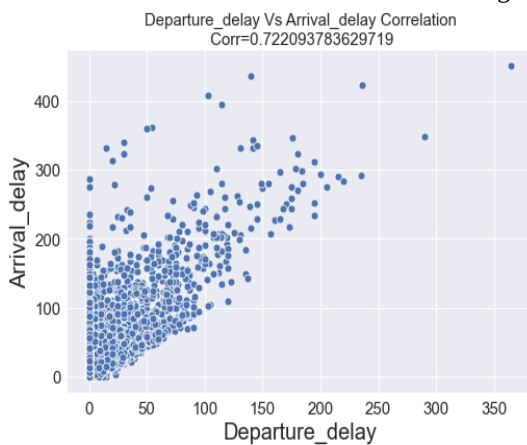


Fig. 4 (a). Departure delay and arriving delay relationship

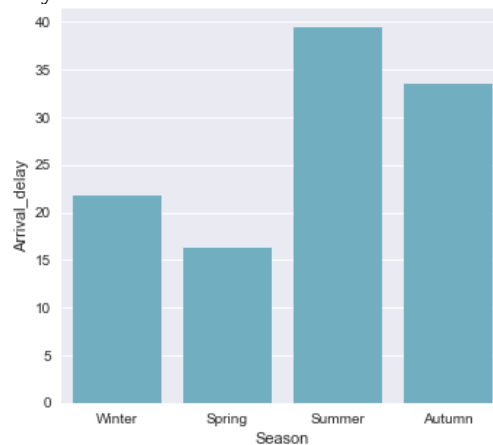


Fig. 4. (b). Seasonal average of delay

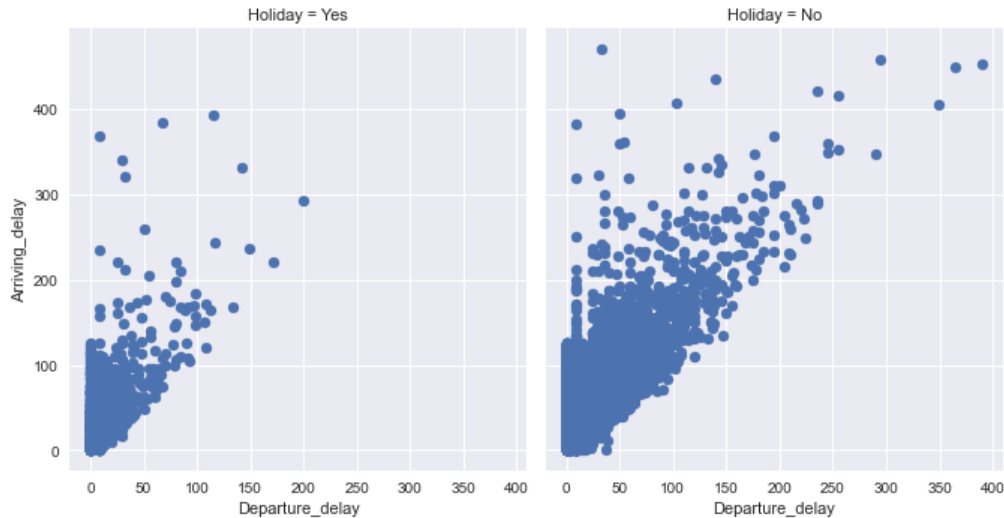


Fig. 5. Relationship between Arrival delay and Departure delay according normal weekdays and holidays

3.4. Feature engineering

After the phase of data analysis and visualization we pass to the feature engineering. Because the model that will be used can only analyze digital data, we converted the categorical columns to numeric value using one-hot-encoding technology. We have used “get_dummies()” function that converted the categorical features to 0 or 1 data by adding additional columns. For example, the Season feature has four values: Winter, Spring, Summer, Autumn. One-hot-encoding technology converted them into binary vectors: [1,0,0,0], [0,1,0,0], [0,0,1,0], [0,0,0,1] respectively. In the same way, the rest of the categorical features are also converted into binary vectors. Now our data is ready for modeling.

3.5. Modeling

Gradient Boosting Decision Tree (GBDT) is a popular, competitive, highly robust, interpretable machine learning algorithm. It is being widely-used in many machine learning tasks and outperform other traditional machine learning models [17]. But with the emergence of big data, the number of both samples and features has been increased. So, in this case the GBDT algorithms facing new challenges, and cannot obtain satisfactory results in term of efficiency and accuracy. When the GBDT algorithms deal with a voluminous data with a high feature dimension the efficiency and scalability are still unsatisfactory. The major reason is that to estimate the information gain of all possible split points for each feature; the GBDT algorithm need to scan all the data instances, which is a very time consuming.

To address these limitations, the LightGBM algorithm [18] was proposed. LightGBM is a gradient boosting framework that uses histogram-based decision tree learning algorithm. It is based on two novel techniques: Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). With GOSS, to reduce the number of data instances, a significant proportion of data instances with small gradients is excluded, and only the rest is used to estimate the information gain. And the EFB, which consist in bundling of exclusive features, to effectively reduce the number of features. It is designed to be an efficient model with multiple advantages, it can handle a large-scale data and achieve a better accuracy with faster training speed and minimum memory usage. It supports parallel and distributed learning.

Benefitting from these advantages, LightGBM is being widely-used in many research areas and has shown very promising results in different machine learning tasks such as: Image classification [19], Speech recognition [20], Social Media Popularity Prediction [21], Text Classification [22], Online Click Fraud Detection [23] ...Additionally to the Lightgbm advantages shown in literature, this algorithm is appropriate in our case, because most of the features in our datasets have a categorical data type, and the number of features has been augmented after the feature engineering step.

In the aim of obtaining more efficient accuracy for the proposed model, we calculated the optimal value for three LightGBM hyperparameters namely: “n_estimators”, “learning_rate” and “num_leaves” using the grid search

algorithm.

- `n_estimators`: number of trees
- `learning_rate`: Models based on gradient boosting create and add new trees to the model sequentially to correct the residual errors in the predictions from the existing sequence of trees. The reason why the model can quickly fit, then overfit the training dataset. The learning rate is applied to slow down the learning as a weighting factor.
- `num_leaves` : represent the max number of nodes per tree.

According to the grid search results, the optimal parameters values for “`n_estimators`”, “`learning_rate`” and “`num_leaves`” parameters that allow us to obtain best performance are : “`n_estimators = 200`”, “`learning_rate = 0.12`” and “`num_leaves = 13`”.

Hence, the processed data was separated into 80 % to training and 20% to test. We used the official python implementation of the Lightgbm model. All the experiments were conducted on I7 3.2GHz 8-core CPU and 16GB of memory. The LightGBM code is available at <https://github.com/Microsoft/LightGBM>

3.6. Evaluation and results

In this section we detailed the evaluation of the proposed methodology and the comparison of obtained results of different models.

3.6.1. Evaluation

To evaluate our method, we compared the obtained results with different machine learning algorithm. The tested models are: Support Vector Regression Model (SVR) [24], Random forest (RFR) [25], Extreme Gradient Boosting (XGboost) [26], Artificial Neural Network (ANN) [27]. The default parameters were used in SVR, RFR and XGboost algorithms and implemented with Python using sklearn library. For neural network is executed using three layers namely input layer, hidden layer and output layer with 100 epochs and it is implemented using the module Keras of the TensorFlow library in Python.

To evaluate the performance of the models, three evaluation factors were used in this study namely R-squared (R^2) (1), The Mean absolute error (MSE) (2), Root Mean Squared Error (RMSE) (3). These statistical parameters are defined as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (1)$$

$$MAE = \frac{1}{N} \sum_{i=1}^n |y_i - \hat{y}| \quad (2)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^n (y_i - \hat{y})^2} \quad (3)$$

Where n denotes the number of target values $y = (y_1, y_2, \dots, y_n)$ and $\hat{y}$ is the predicted value of y .

3.6.2. Results

The following table presents the obtained results for the Lightgbm compared with the tested algorithms. As shown in the table the Lightgbm outperform all tested model and is faster one in running.

Table 2. Comparison of obtained results.

Models	R-squared	RMSE	MAE	Time Taken
SVM	70.483994	23.381866	11.496802	4.226806
RF	79.578848	19.448696	10.733235	2.674848
LGBM	83.362681	17.554645	9.905543	0.050862
XGB	77.666394	20.339009	10.685415	0.533575
ANN	80.597111	15.836527	11.150983	26.945896

The following figures are helpful to better understand and present the final results. As shown in the following figures the proposed method has the superior score in R_squared reaches more than 83 % in term of R-squared where it is outperformed the RF, SVR, XGBoost and NN. In addition, it has the minimum score in RMSE, MAE and it was the faster one with the minimum time for training process. The results of LighGBM and Neural Network were similar but the LightGBM is the faster one in running time.

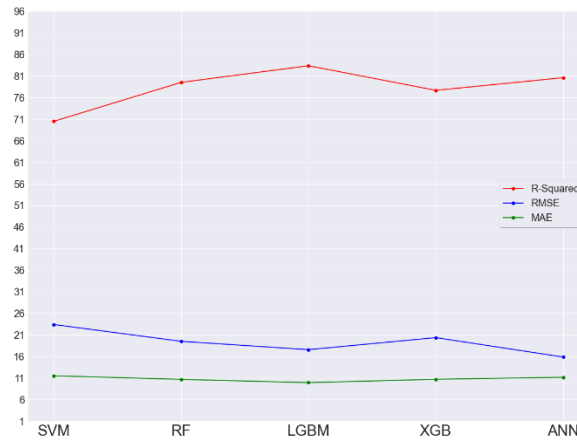


Fig. 6. Score comparison

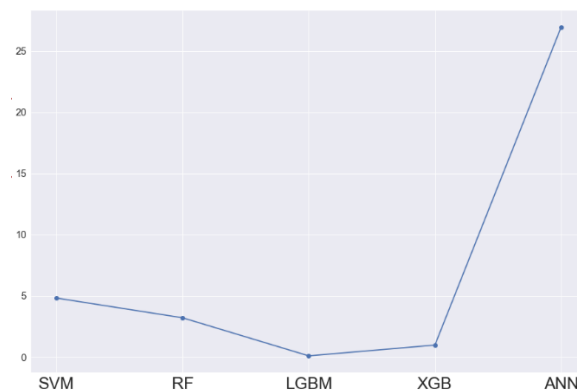


Fig. 7. Time running (second) comparison

4. Conclusion

This study presents a detailed predictive analysis process based on machine learning algorithm to analyze the relationship between train delays and different characteristics of Tunisian railway system. In this study, the LightGBM model is proposed to construct a prediction model for train arrival delay, and the model is trained and tested based on the historical data of train operation collected by the SNCFT and infrastructure characteristics. Our method presents the first application of LightGBM algorithm in the train delay prediction task and also the first work that address the prediction of train delays in Tunisia. Our methodology consisted of the following phases: collection, cleaning, full analyzing, feature engineering and modeling and we have detailed each phase. The results show that the LightGBM is more powerful, faster, and greatly improved in performance. As indicated by the data validation results, our proposed method based on LightGBM algorithms outperform all the other tested algorithms in term of both accuracy and time running. Specifically, the LightGBM model exhibits better MAE and MSE values, and its prediction accuracy reaches 83 % within 0.05 s. For our future works we will integrate other extern influence factors that can affect the delay in railway systems such as weather data to obtain better prediction results.

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